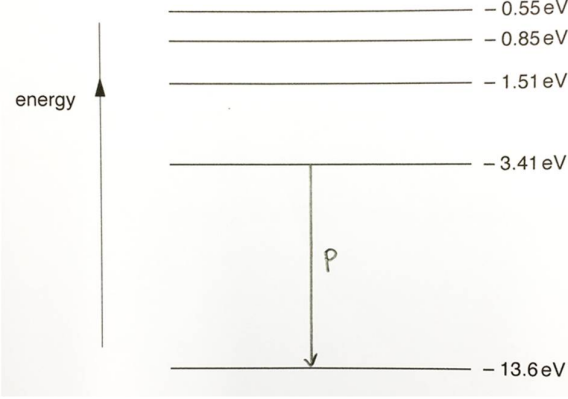


	$M' \approx 0.0533 \text{ kg}$
8(a)	A photon is a quantum of electromagnetic radiation with energy $E = hf$, where h is the planck constant and f is the frequency of the radiation.
8(b)(i)	There are discrete coloured lines on a dark background, where red is the longer wavelength and violet is the shortest wavelength.
8(b)(ii)	The discrete coloured lines represent the wavelengths or energy equivalence of the photons emitted during de-excitation between energy levels of a gas. Since only specific lines are observables, it means that only photons of certain energy can be emitted. Hence, this shows that the energy levels in an atom are discrete and not continuous.
8(c)(i)	Energy of a photon of violet light, $E = \frac{hc}{\lambda}$ $E = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{400 \times 10^{-9}}$ $E = 4.9725 \times 10^{-19} \text{ J}$ $E \approx 3.11 \text{ eV}$ The smallest energy transition to the -13.6 eV is from the -3.41 eV, which resulted in a photon of energy 10.19 eV being emitted. This far exceeds the energy of a photon of violet light. A photon of red light, which has a smaller wavelength, would have a smaller energy than 3.11 eV. Hence, the electron transitions to the energy level -13.6 eV do not result in emissions in the visible spectrum.
8(c)(ii)1.	3 lines.

8(c)(ii)2.	The shortest wavelength corresponds to the largest wavelength in the visible spectrum. $E = \frac{hc}{\lambda}$ $(3.41 - 0.55) \times 1.60 \times 10^{-19} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{\lambda}$ $\lambda \approx 435 \text{ nm}$
8(d)(i)	
8(d)(ii)	Using the photoelectric effect equation, $E = \phi + K.E._{\text{max}}$ $13.6 - 3.41 = 5.6 + K.E._{\text{max}}$ $K.E._{\text{max}} = 4.59 \text{ eV}$
8(e)(i)	$c = \frac{d}{t}$ $3.0 \times 10^8 = \frac{2.6 \times 10^{-10}}{t}$ $t \approx 8.67 \times 10^{-19} \text{ s}$
8(e)(ii)	Maximum uncertainty, $\Delta t = 8.67 \times 10^{-19} \text{ s}$
8(e)(iii)	By Heisenberg's uncertainty principle, $\Delta E \Delta t \geq \frac{\hbar}{2}$ $\Delta E \times 8.67 \times 10^{-19} \geq \frac{6.63 \times 10^{-34}}{4\pi}$ $\Delta E \geq 6.09 \times 10^{-17} \text{ J}$ As the photon crosses the atom, there is an uncertainty in time taken to cross the gold atom and a corresponding uncertainty in its energy. Hence, its energy will not be equal to that of a photon in free space.
9(a)(i)	$a = \frac{GM}{r^2} \text{ (magnitude)}$